JAVA Programming Project Presentation

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- Project Title: Restaurent order processing and billing system
- Course: Java Mini Project
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Introduction

What is a Restaurant Management System?

A Restaurant Management System is a software application that helps manage restaurant
operations like taking orders, processing them, and generating bills automatically.

- Purpose of the Project:

The purpose of this project is to automate the ordering and billing process, reduce human errors, and make restaurant operations faster and more efficient.

- Importance in Real-Life Restaurants:

Saves time by speeding up order and billing processReduces mistakes in orders and calculations
Improves customer serviceHelps manage records easily

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Objective

- Automate restaurant order management.
- Reduce manual billing errors.
- Provide fast and accurate bill generation.
- Improve customer service efficiency.
- Maintain a structured record of orders.

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Problem Statement

Manual order processing in restaurants can lead to:

- Incorrect billing
- Delays in service
- Difficulty tracking orders
- Human errors in calculations

This system solves these problems by digitizing order entry and billing.

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System Requirements

Hardware Requirements
Computer / Laptop
Minimum 4GB RAM

Software Requirements
Java (JDK 8 or above)
Any IDE (Eclipse / IntelliJ / VS Code)

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System Design

- Menu Management
- Displays available food items and prices.
- Order Processing
- Takes customer orders.
- Stores item quantities.
- Billing System
- Calculates total cost.
- Applies tax if needed.
- Receipt Generation
- Prints formatted bill.

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Module Description

Admin Module: Manages menu items and controls the system.
Order Processing Module: Handles customer orders and item selection.
Billing Module: Calculates total and generates the bill.
Customer Module: Stores and manages customer details.
Payment Module: Processes and confirms payments.

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Algorithm / Flowchart

- Start
- ↓
- Enter Savings Goal
- ↓
- Enter Current Savings
- ↓
- Calculate Remaining Amount
- ↓
- Calculate Percentage Achieved
- ↓
- Display Results
- ↓
- End

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Implementation

```
import java.util.Scanner;

public class Billing {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int choice, qty;
        double total = 0;

        System.out.println("1.Burger ₹100  2.Pizza ₹200  3.Juice ₹50");

        System.out.print("Enter choice: ");
        choice = sc.nextInt();

        System.out.print("Enter quantity: ");
        qty = sc.nextInt();

        switch(choice) {
            case 1: total = qty * 100; break;
            case 2: total = qty * 200; break;
            case 3: total = qty * 50; break;
            default: System.out.println("Invalid choice");
        }

        double tax = total * 0.05;
        double bill = total + tax;

        System.out.println("Total = ₹" + total);
        System.out.println("Final Bill = ₹" + bill);
    }
}
```

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Output Screens

```
--- MENU ---
1. Burger - ₹100
2. Pizza - ₹200
3. Juice - ₹50
4. Exit
Enter your choice: 1
Enter quantity: 2

--- MENU ---
Enter your choice: 3
Enter quantity: 1

--- MENU ---
Enter your choice: 4

Total Amount: ₹250
Tax (5%): ₹12.5
Final Bill: ₹262.5
```

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Results & Discussion

The system successfully processes orders and generates bills.
Accurate calculation of total amount and tax.
Reduces manual errors and saves time.
Easy to use and understand.

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Conclusion & Future Enhancement

The system effectively helps users plan and track their savings goals. It simplifies financial tracking and promotes better money management habits. This basic system can be further enhanced with features like data storage, multiple goals, and graphical dashboards.

- Future Enhancements:
- • Add GUI support
- • Integrate with database
- • Extend functionality for scalability

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References / Acknowledgement

1. Java: The Complete Reference – For understanding Java programming concepts.
2. Head First Java – For beginner-friendly Java learning.
3. Oracle Java Documentation
4. https://docs.oracle.com/javase/8/docs/ – Official Java documentation for syntax and APIs.
5. GeeksforGeeks
6. https://www.geeksforgeeks.org/java/ – Tutorials and examples on Java programming.
7. W3Schools Java Tutorial
8. https://www.w3schools.com/java/ – Basic Java concepts and examples.
9. Personal Finance Management – General principles of saving and budgeting.

Thank You